

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$y = ax^2 - c$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy-plane?

Answer

- A. Zero
- B. One
- C. Two
- D. More than two

Correct Answer: C

Rationale

Choice C is correct. It is given that the constants a and c are both positive; therefore, the graph of the given quadratic equation is a parabola that opens up with a vertex on the y-axis at a point below the x-axis. The graph of the second equation provided is a horizontal line that lies above the x-axis. A horizontal line above the x-axis will intersect a parabola that opens up and has a vertex below the x-axis in exactly two points.

Choices A, B, and D are incorrect and are the result of not understanding the relationships of the graphs of the two equations given. Choice A is incorrect because the two graphs intersect. Choice B is incorrect because in order for there to be only one intersection point, the horizontal line would have to intersect the parabola at the vertex, but the vertex is below the x-axis and the line is above the x-axis. Choice D is incorrect because a line cannot intersect a parabola in more than two points.